

ADLink: Multi-Channel Advertising Management Platform

Database Design Document

Version 1.0

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Chapter 1

Project Overview

1.1 Introduction

Advertising is a key part of modern business, but managing campaigns across multiple platforms can be time-consuming and complex. Businesses often rely on different agencies and services, which makes coordination difficult. To address this, we propose a Multi Channel Advertising Management Platform that consolidates all advertising options into a single digital system.

The platform enables business owners to manage campaigns from one dashboard, allowing them to book services such as online promotions, social media marketing, billboard advertising, and vehicle branding. Business owners can also reserve billboard spaces in various cities across Pakistan through an organized interface.

Using a centralized database, the system efficiently stores and manages information on users, advertising services, billboard locations, bookings and payments. This ensures streamlined data management, transparent pricing, and easy campaign tracking.

Developed under ADLink, the platform makes professional marketing accessible to large businesses, startups, and small online store owners, simplifying the advertising process and improving operational efficiency.

1.2 Problem Statement

The current advertising system in Pakistan has several major problems that make it difficult to manage campaigns and advertise businesses effectively.

- **Lack of Integration:** Business owners have to deal with billboard owners, vehicle branding companies and advertisers separately, usually through informal communication, because there is no single platform that connects them all.
- **Lack of Transparency:** There is no proper centralized system to find advertising services, check prices, see availability, or confirm bookings, which makes the advertising process hectic and complex.

- **Manual Processes:** Campaign tracking, payments, and bookings are often done manually using spreadsheets or phone calls, which keeps the data separate and hard to manage.
- **No Centralized Booking:** Without a structured booking system, double-bookings, cancellations, and unresolved disputes are common, damaging business relationships.

ADLYNK DIRECTLY SOLVES THESE PROBLEMS

ADLink provides a centralized platform that brings billboard owners, advertisers, business owners and vehicle branding companies, together in one system. Businesses can easily discover available advertising services, view prices, and check details, while bookings and payments are managed securely within the platform. All campaigns, services, and transactions are stored in an organized database, reducing manual work and inefficiencies. This streamlined approach allows businesses to focus on improving their products, services, and growth, rather than dealing with the complexities of managing advertising campaigns.

1.3 Objectives

The main objectives of the ADLink Multi-Channel Advertising Management Platform are:

1. To integrate all advertising services in one platform, so business owners no longer need to contact billboard owners, vehicle branding companies, and advertisers
2. To provide transparency in service discovery, pricing, availability, and booking confirmation, reducing a hectic and complex advertising process. separately.
3. To allow businesses to book and manage campaigns directly through the platform, reducing manual effort and complexities.
4. To automate campaign tracking, payments, and bookings, eliminating manual spreadsheets and phone call processes.
5. To establish a centralized booking system that prevents double-bookings, cancellations, and unresolved disputes.
6. To allow businesses to focus on improving products, services, and scaling, rather than managing advertising complexities.
7. To provide a centralized platform where businesses—including small e-commerce stores—can easily access high-quality advertising services, manage campaigns, and make secure bookings, while service providers can reach clients efficiently, ensuring better services and smooth operations for everyone.

1.4 GitHub Repository Link

https://github.com/bscs24f48-sys/Software_Engineeringproject.git

Chapter 2

Detailed Database Design

2.1 Entities

Sr. No	Entity Name	Description
01	USER	Stores shared information for all users registered on the platform.
02	BUSINESS_OWNER	Represents clients who create campaigns and purchase advertising services.
03	ASSET_OWNER	Represents users who own physical advertising assets such as billboards and vehicles.
04	ADVERTISER	Represents users providing digital advertising services.
05	BILLBOARD	Stores information about billboard advertising assets.
06	VEHICLE	Stores information about vehicles available for advertising.
07	DIGITAL_SERVICE	Represents online advertising and marketing services.
08	CAMPAIGN	Represents advertising campaigns created by business owners.
09	BOOKING	Represents reservations of advertising services or assets.
10	PAYMENT	Stores payment transaction details for bookings.
11	WITHDRAWAL	Represents withdrawal requests made by service providers.

12	REVIEW	Stores feedback and ratings submitted by users.
13	ACCOUNT_BALANCE	Stores financial balance information for advertisers and asset owners.
14	LEDGER_ENTRY	Maintains complete transaction history and audit records for balance changes.

2.2 Data Dictionary

2.2.1 USER (Supertype)

Sr. No	Name	Data Type	Constraint	Description
01	user_id (PK)	INTEGER	NOT NULL	Unique identifier for every user
02	name	VARCHAR(100)	NOT NULL	Full name of the user
03	email	VARCHAR(100)	UNIQUE, NOT NULL	User email address
04	password_hash	VARCHAR(255)	NOT NULL	Encrypted password value
05	phone	VARCHAR(20)	UNIQUE	Contact number of the user
06	city	VARCHAR(100)	NULL	User city location
07	bio	TEXT	NULL	User profile description
08	role	VARCHAR(30)	NOT NULL	Defines user type in the system
09	created_at	DATETIME	NOT NULL	Account creation timestamp
10	updated_at	DATETIME	NOT NULL	Last account update timestamp

2.2.2 BUSINESS_OWNER

Sr. No	Name	Data Type	Constraint	Description
01	owner_id (PK)	INTEGER	NOT NULL	Unique business owner identifier

02	user_id (FK)	INTEGER	NOT NULL	References USER(user_id)
03	company_name	VARCHAR(100)	NOT NULL	Registered business name
04	industry	VARCHAR(50)	NULL	Business industry category

2.2.3 ASSET_OWNER

Sr. No	Name	Data Type	Constraint	Description
01	asset_owner_id (PK)	INTEGER	NOT NULL	Unique asset owner identifier
02	user_id (FK)	INTEGER	NOT NULL	References USER(user_id)
03	company_name	VARCHAR(100)	NOT NULL	Asset owner company name
04	bank_account	VARCHAR(50)	NOT NULL	Registered bank account number
05	cnic	VARCHAR(20)	UNIQUE, NOT NULL	National identity card number

2.2.4 ADVERTISER

Sr. No	Name	Data Type	Constraint	Description
01	advertiser_id (PK)	INTEGER	NOT NULL	Unique advertiser identifier
02	user_id (FK)	INTEGER	NOT NULL	References USER(user_id)
03	agency_name	VARCHAR(100)	NOT NULL	Name of advertising agency
04	specialization	VARCHAR(100)	NULL	Area of digital marketing expertise

2.2.5 BILLBOARD

Sr. No	Name	Data Type	Constraint	Description
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01	billboard_id (PK)	INTEGER	NOT NULL	Unique billboard identifier
02	asset_owner_id (FK)	INTEGER	NOT NULL	References ASSET_OWNER
03	title	VARCHAR(100)	NOT NULL	Billboard title
04	city	VARCHAR(100)	NOT NULL	City where billboard exists
05	location_address	VARCHAR(255)	NOT NULL	Billboard location address
06	dimensions	VARCHAR(50)	NOT NULL	Billboard size dimensions
07	price_per_day	DECIMAL(10,2)	NOT NULL	Daily rental cost
08	availability_status	VARCHAR(20)	NOT NULL	Availability status
09	created_at	DATETIME	NOT NULL	Billboard listing timestamp

2.2.6 VEHICLE

Sr. No	Name	Data Type	Constraint	Description
01	vehicle_id (PK)	INTEGER	NOT NULL	Unique vehicle identifier
02	asset_owner_id (FK)	INTEGER	NOT NULL	References ASSET_OWNER
03	title	VARCHAR(100)	NOT NULL	Vehicle advertisement title
04	vehicle_type	VARCHAR(50)	NOT NULL	Type of vehicle
05	fleet_size	INTEGER	NOT NULL	Number of vehicles available
06	route_coverage	VARCHAR(100)	NULL	Vehicle operational route area
07	city	VARCHAR(100)	NOT NULL	City of operation
08	price_per_day	DECIMAL(10,2)	NOT NULL	Daily advertising price
09	availability_status	VARCHAR(20)	NOT NULL	Vehicle availability state

10	created_at	DATETIME	NOT NULL	Vehicle listing timestamp
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2.2.7 DIGITAL_SERVICE

Sr. No	Name	Data Type	Constraint	Description
01	service_id (PK)	INTEGER	NOT NULL	Unique service identifier
02	advertiser_id (FK)	INTEGER	NOT NULL	References ADVERTISER
03	title	VARCHAR(100)	NOT NULL	Digital service title
04	description	TEXT	NULL	Service details description
05	platform	VARCHAR(50)	NOT NULL	Advertising platform name
06	price	DECIMAL(10,2)	NOT NULL	Service package price
07	availability_status	VARCHAR(20)	NOT NULL	Service availability status
08	created_at	DATETIME	NOT NULL	Service creation timestamp

2.2.8 CAMPAIGN

Sr. No	Name	Data Type	Constraint	Description
01	campaign_id (PK)	INTEGER	NOT NULL	Unique campaign identifier
02	owner_id (FK)	INTEGER	NOT NULL	References BUSINESS_OWNER(owner_id)
03	title	VARCHAR(100)	NOT NULL	Campaign title
04	description	TEXT	NULL	Campaign details and objectives
05	status	VARCHAR(20)	NOT NULL	Campaign current status
06	start_date	DATE	NOT NULL	Campaign start date

07	end_date	DATE	NOT NULL	Campaign ending date
08	total_budget	DECIMAL(10,2)	NOT NULL	Total campaign allocated budget
09	created_at	DATETIME	NOT NULL	Campaign creation timestamp

2.2.9 BOOKING

Sr. No	Name	Data Type	Constraint	Description
01	booking_id (PK)	INTEGER	NOT NULL	Unique booking identifier
02	owner_id (FK)	INTEGER	NOT NULL	References BUSINESS_OWNER(owner_id)
03	campaign_id (FK)	INTEGER	NOT NULL	References CAMPAIGN(campaign_id)
04	booking_type	VARCHAR(30)	NOT NULL	Type of booking asset or service
05	billboard_id (FK)	INTEGER	NULL	References BILLBOARD(billboard_id)
06	vehicle_id (FK)	INTEGER	NULL	References VEHICLE(vehicle_id)
07	service_id (FK)	INTEGER	NULL	References DIGITAL_SERVICE(service_id)
08	num_vehicles	INTEGER	NULL	Number of vehicles booked
09	status	VARCHAR(20)	NOT NULL	Booking current status
10	start_date	DATE	NOT NULL	Booking start date
11	end_date	DATE	NOT NULL	Booking ending date
12	total_amount	DECIMAL(10,2)	NOT NULL	Total booking payment amount
13	booked_at	DATETIME	NOT NULL	Booking creation timestamp

2.2.10 PAYMENT

Sr. No	Name	Data Type	Constraint	Description
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01	payment_id (PK)	INTEGER	NOT NULL	Unique payment identifier
02	booking_id (FK)	INTEGER	NOT NULL	References BOOKING(booking_id)
03	business_owner_id (FK)	INTEGER	NOT NULL	References BUSINESS_OWNER(owner_id)
04	amount	DECIMAL(10,2)	NOT NULL	Total payment amount
05	platform_fee	DECIMAL(10,2)	NOT NULL	Platform service fee amount
06	provider_amount	DECIMAL(10,2)	NOT NULL	Amount transferred to provider
07	status	VARCHAR(20)	NOT NULL	Payment transaction status
08	payment_method	VARCHAR(50)	NOT NULL	Payment method used
09	paid_at	DATETIME	NOT NULL	Payment completion timestamp

2.2.11 WITHDRAWAL

Sr. No	Name	Data Type	Constraint	Description
01	withdrawal_id (PK)	INTEGER	NOT NULL	Unique withdrawal request identifier
02	requester_id (FK)	INTEGER	NOT NULL	References USER(user_id)
03	requester_type	VARCHAR(30)	NOT NULL	Type of requester account
04	amount	DECIMAL(10,2)	NOT NULL	Requested withdrawal amount
05	status	VARCHAR(20)	NOT NULL	Withdrawal processing status
06	bank_account	VARCHAR(50)	NOT NULL	Bank account used for transfer
07	requested_at	DATETIME	NOT NULL	Withdrawal request timestamp
08	processed_at	DATETIME	NULL	Withdrawal completion timestamp

2.2.12 REVIEW

Sr. No	Name	Data Type	Constraint	Description
01	review_id (PK)	INTEGER	NOT NULL	Unique review identifier
02	booking_id (FK)	INTEGER	NOT NULL	References BOOKING(booking_id)
03	reviewer_id (FK)	INTEGER	NOT NULL	References USER(user_id)
04	reviewee_id (FK)	INTEGER	NOT NULL	References USER(user_id)
05	rating	INTEGER	CHECK (rating BETWEEN 1 AND 5)	Numerical review rating
06	comment	TEXT	NULL	Written feedback comment
07	created_at	DATETIME	NOT NULL	Review submission timestamp

2.2.13 ACCOUNT_BALANCE

Sr. No	Name	Data Type	Constraint	Description
01	balance_id (PK)	INTEGER	NOT NULL	Unique balance identifier
02	user_id (FK)	INTEGER	NOT NULL	References USER
03	user_type	VARCHAR(30)	NOT NULL	Type of provider account
04	current_balance	DECIMAL(10,2)	NOT NULL	Current available balance
05	pending_balance	DECIMAL(10,2)	NOT NULL	Pending balance amount
06	last_updated	DATETIME	NOT NULL	Timestamp of latest balance update

2.2.14 LEDGER_ENTRY

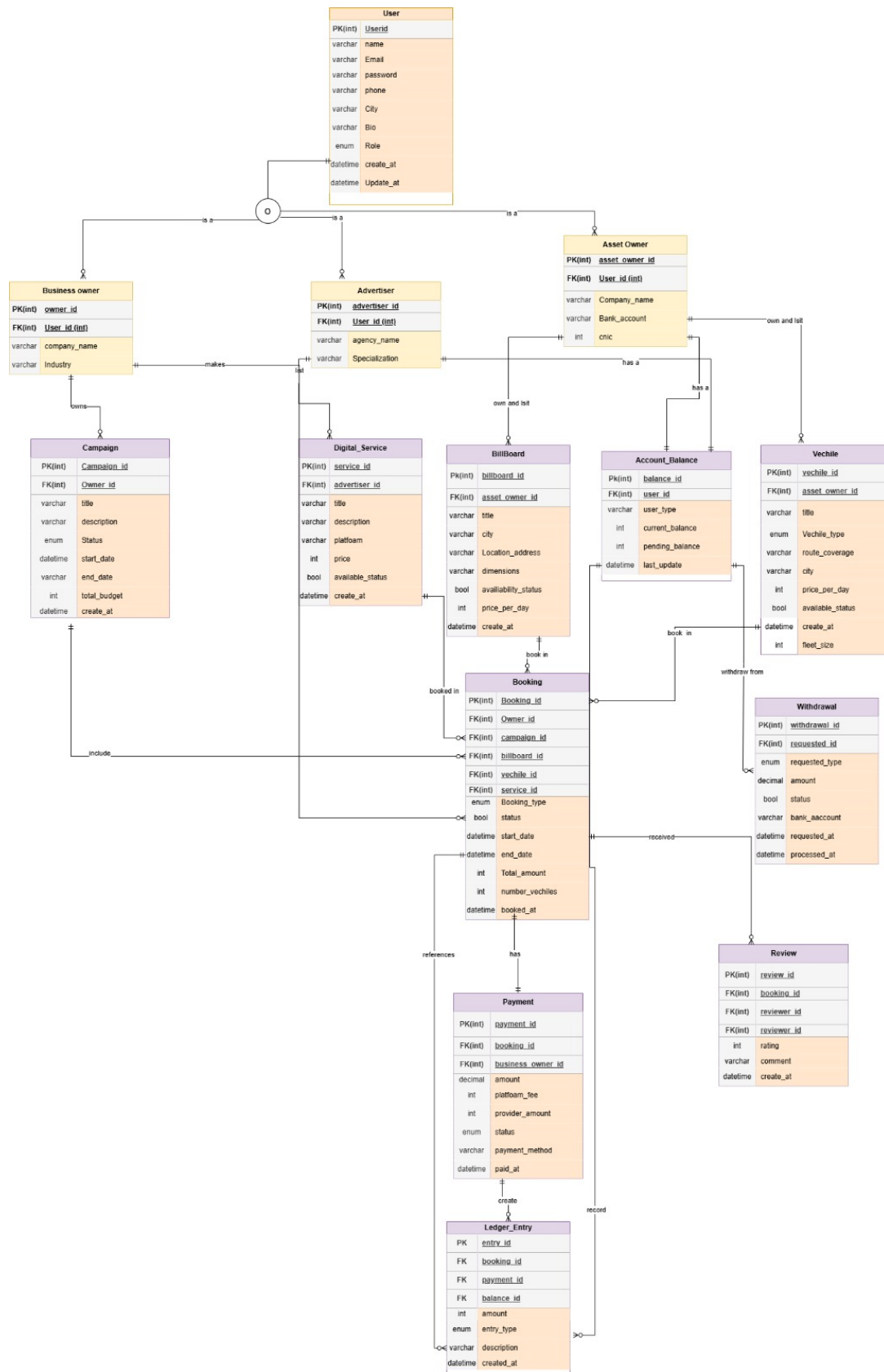
Sr. No	Name	Data Type	Constraint	Description
01	entry_id (PK)	INTEGER	NOT NULL	Unique ledger entry identifier
02	balance_id (FK)	INTEGER	NOT NULL	References ACCOUNT_BALANCE
03	payment_id (FK)	INTEGER	NULL	References PAYMENT
04	booking_id (FK)	INTEGER	NULL	References BOOKING
05	amount	DECIMAL(10,2)	NOT NULL	Amount credited or debited
06	entry_type	VARCHAR(20)	NOT NULL	Indicates credit or debit transaction
07	description	TEXT	NULL	Description of ledger activity
08	created_at	DATETIME	NOT NULL	Date and time of entry creation

2.3 Business Rules

1. A USER can be one or more roles (BUSINESS_OWNER, ADVERTISER, ASSET_OWNER) simultaneously but must have at least one role to use the platform. BUSINESS_OWNER can only buy services/assets, ADVERTISER can only provide digital services, and ASSET_OWNER can only own physical assets.
2. An ASSET_OWNER can list multiple BILLBOARDS and VEHICLES but each asset belongs to exactly one owner. Only assets with ‘‘available’’ status can be booked and an asset cannot be double-booked for overlapping dates.
3. An ADVERTISER can list multiple DIGITAL_SERVICES but each service belongs to exactly one advertiser. Service price is fixed per campaign.
4. A BUSINESS_OWNER can own multiple CAMPAIGNS. Each CAMPAIGN contains exactly one BOOKING and each BOOKING belongs to exactly one CAMPAIGN. CAMPAIGN.total_budget equals BOOKING.total_amount and a CAMPAIGN cannot be deleted if it has an associated BOOKING.
5. A BUSINESS_OWNER can make multiple BOOKINGS. Each BOOKING must reference exactly one of billboard_id, vehicle_id, or service_id based on booking_type while the other two remain NULL. Booking dates cannot overlap for the same asset and VEHICLE bookings cannot exceed fleet size.
6. A BOOKING has exactly one PAYMENT and a PAYMENT belongs to exactly one BOOKING. PAYMENT.amount must equal BOOKING.total_amount and payment must be completed before provider credit is issued.

7. Every ADVERTISER and ASSET_OWNER has exactly one ACCOUNT_BALANCE. current_balance equals the SUM of all related LEDGER_ENTRY.amount values.
8. Every balance change must create a LEDGER_ENTRY before updating ACCOUNT_BALANCE. Positive amounts represent credits while negative amounts represent debits. LEDGER_ENTRY records cannot be deleted or modified.
9. Only ADVERTISERS and ASSET_OWNERS can request WITHDRAWALS. Withdrawal amount cannot exceed ACCOUNT_BALANCE.current_balance and users cannot create a new withdrawal request while another is pending.
10. A BOOKING can have at most two REVIEWS. Reviews can only be submitted after booking completion and rating must remain between 1 and 5.
11. Platform_fee is a fixed percentage of PAYMENT.amount. Providers can withdraw only after booking start date and all monetary values are stored in PKR.

2.4 Entity Relationship Diagram



References

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Appendix: AI Usage and Prompts

AI Model	Prompt Purpose	Prompt Text
Gemini	Structural Compliance	‘‘I am submitting Milestone 2 for a Database Systems course. Review my Introduction and Problem Statement against the Namal University template guidelines. Ensure the tone is academic and focuses on the real-world advertising environment in Pakistan.’’
Gemini	Objective Refinement	‘‘We have identified five specific objectives for ADLink. Can you help us rephrase them so they sound more professional and focus on automation and centralization of advertising workflows?’’
Gemini	General Overview	‘‘Explain how a centralized platform like ADLink specifically solves the issue of informal communication in the Pakistani billboard industry so we can include it in our report logic.’’
Claude	Technical Constraint Audit	‘‘In our EERD, we have a User supertype with Business Owner and Service Provider subtypes. If we want to ensure no user can accidentally have both roles, should we use a Disjoint or Overlapping constraint? Please provide the technical justification.’’

Claude	Relationship Mapping	‘‘Review the relationship between Campaign and Booking. If one campaign can contain multiple billboard and digital bookings, how should we represent the cardinality and participation to ensure data integrity?’’
Claude	EERD Logic Verification	‘‘Look at our list of entities: [User, Business Owner, Asset Owner, Advertiser, Billboard, Vehicle, Digital Service, Campaign, Payment, Withdrawal, Review]. Are there any missing entities that would be required to handle a multi-channel booking lifecycle?’’
GPT-4o	Domain Attribute Research	‘‘What are the standard technical attributes required for a Billboard entity in a commercial advertising database? We need to include things like location coordinates, daily traffic reach, and illumination status.’’
GPT-4o	Data Type Validation	‘‘Check our Data Dictionary attributes for the Payment and Withdrawal entities. Are Decimal and Integer types appropriate for currency and tracking numbers in a standard SQL database?’’
GPT-4o	Final Document Audit	‘‘Acting as a Database Professor, audit our Problem Statement. Does it clearly identify a need for a database-driven solution, or is it too focused on general business problems? Suggest one improvement.’’